CSS

Position, pseudo class, pseudo element, float

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Position</title>
    <style>
      /* h1{
        position:relative;
        top: 100px;
      } */

      h2 {
        position: absolute;
        z-index: 1001;
        background-color: blueviolet;
        top: 100px;
      }
      div {
        /* position: relative; */
      }
      h3 {
        position: fixed;
        top: 100px;
        z-index: 1000;
        background-color: aqua;

      }
      h4 {
        position: sticky;
        bottom: 0;
      }

      img {
        width: 100px;
        height: 100px;
      }
      header {
        position: sticky;
        top: 0;
      }
    </style>
```

```html
  </head>
  <body>
    <!-- <h1>Position</h1> -->

    <!-- position => static, relative, absolute, fixed , sticky -->

    <h1>Relative element</h1>
    <div>
      <p>Hello there</p>
      <h2>Absolute element</h2>
    </div>
    <h3>Fixed element</h3>
    <h4>Sticky element</h4>
    <!-- <h5>below</h5> -->

    <header>
      <img
        src="https://www.boat-lifestyle.com/cdn/shop/files/Artboard1_2d0001fd-
30bb-4390-89bf-cab91a163c8a_900x.png?v=1759823717"
        alt=""
      />
    </header>

    <p>
      Lorem ipsum dolor sit amet consectetur, adipisicing elit. Temporibus
illum
      distinctio unde provident laudantium natus soluta pariatur deleniti,
      repudiandae qui quasi veritatis nostrum recusandae. Neque sit aliquid
      quisquam ex id! Explicabo fugit facere dolorem tempore iste vero, omnis
      ratione. Nostrum, at. Tempore architecto assumenda odio ad aliquam
    </p>
  </body>
</html>
```

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Pseudo</title>

    <style>
      /* : colon => psuedo class  */

      h1:hover {
        background-color: yellow;
```

```
    border: 20px solid black;
    font-size: 50px;
  }

  a:link {
    color: aqua;
  }

  a:visited {
    color: green;
  }

  a:active {
    color: black;
  }
  input:focus {
    outline-color: red;
  }

  /* .ul li:first-child {
    color: blue;
  }

  li:last-child {
    color: yellow;
  }

  li:nth-child(odd) {
    color: red;
  } */

  li::after {
    content: "😊";
  }

  li::before {
    content: "😍";
  }

  p::first-letter {
    text-transform: capitalize;
    font-size: 50px;
  }

  p::first-line {
    text-transform: capitalize;
  }
```

```
    ::selection {
      background-color: brown;
      color: white;
    }

    ::marker {
      color: green;
      font-size: 100px;
    }

    .container > div {
      width: 100px;
      height: 100px;
      background-color: tomato;
      margin-right: 10px;
      float: right;
    }
  </style>
</head>
<body>
  <div class="container">
    <div></div>
    <div></div>
    <div></div>
  </div>

  <p>
    lorem ipsum dolor sit amet consectetur adipisicing elit. Fuga ipsa,
    deleniti quae delectus doloremque nisi quidem in aperiam iusto, tenetur
    provident optio, exercitationem sequi! Aspernatur iure doloremque
    dignissimos eaque culpa!
  </p>
  <ul class="ul">
    <li>item 1</li>
    <li>item 2</li>
    <li>item 3</li>
    <li>item 4</li>
    <li>item 5</li>
    <li>item 6</li>
    <li>item 7</li>
    <li>item 8</li>
    <li>item 9</li>
    <li>item 10</li>
  </ul>

  <ul>
    <li>item 1</li>
    <li>item 2</li>
```

```html
      <li>item 3</li>
      <li>item 4</li>
      <li>item 5</li>
      <li>item 6</li>
      <li>item 7</li>
      <li>item 8</li>
      <li>item 9</li>
      <li>item 10</li>
    </ul>
    <a href="#">Google</a>
    <a href="#data">Atlassian</a>
    <input type="text" />

    <h1>Pseudo</h1>
  </body>
</html>
```

Theory Notes

1. CSS Positioning

- **Static** (default): Elements are positioned in normal document flow.
- **Relative**: Positioned relative to its normal position. Useful for small shifts.
- **Absolute**: Positioned relative to the nearest positioned ancestor (`relative`, `absolute`, `fixed`, `sticky`). Removed from normal flow.
- **Fixed**: Positioned relative to the viewport. Stays in place when scrolling.
- **Sticky**: Acts like relative until a scroll threshold is reached, then sticks to a position.

z-index: Controls stacking order. Higher values appear above lower ones.

2. Pseudo-classes

- **Definition**: Keywords added to selectors to style elements in specific states.
- Examples:
  - `:hover` → when mouse hovers.
  - `:active` → when clicked.
  - `:visited` → visited links.
  - `:focus` → when input is focused.
  - `:first-child`, `:last-child`, `:nth-child(n)` → target specific children.

3. Pseudo-elements

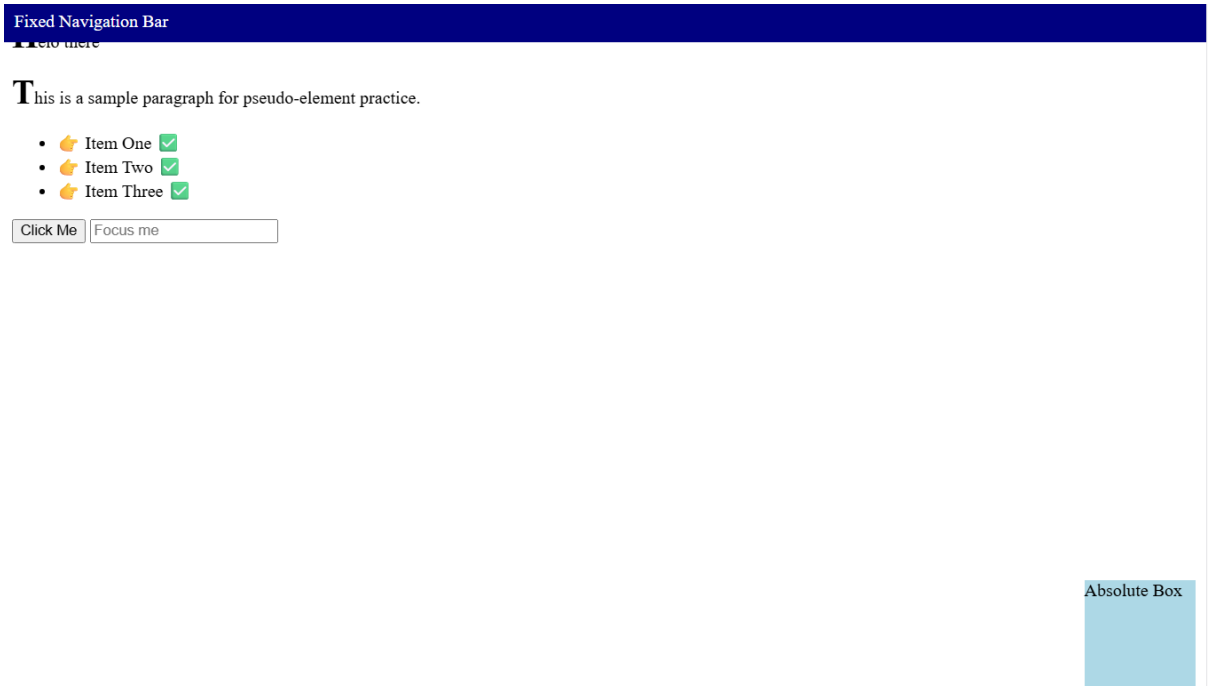
- **Definition**: Allow styling specific parts of an element.
- Examples:
  - `::before` → insert content before element.
  - `::after` → insert content after element.

- `::first-letter` → style first letter.
- `::first-line` → style first line.
- `::selection` → style text when selected.
- `::marker` → style list markers.

4. Float

- `float: left/right;` → pushes elements to one side, allowing text/content to wrap around.
- Often replaced by Flexbox/Grid in modern layouts, but still useful for simple cases.

Practice Tasks

Coding Exercises

1. Create a navigation bar that stays fixed at the top of the page using `position: fixed`.
2. Use `position: absolute` to place a box inside a container at the bottom-right corner.
3. Style a paragraph so that the first letter is larger and bold using `::first-letter`.
4. Add emojis before and after each list item using `::before` and `::after`.
5. Create a button that changes background color when hovered and becomes black when clicked (`:hover`, `:active`).
6. Style input fields so that their border turns red when focused.
7. Use `::selection` to change the highlight color when selecting text.